

FH Aachen

Department of Electrical Engineering and Information Technology

Exposé

**Evaluating AI Agent Capabilities in SysML v2 Modeling
via an MCP Bridge to Magic Systems of Systems Architect**

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Master's Thesis – WS 2026/2027
Aachen, September 2026

1 Exposé

Motivation

[TODO 150 words]

Problem Statement

[TODO 150 words]

Research Question and Hypotheses

RQ: Which SysML v2 modeling tasks can an LLM-based agent perform reliably in an industrial modeling environment (MSoSA) when connected through an MCP bridge, and where do its capabilities break down?

- **H1 (Task asymmetry):** Read-only tasks (Abfragen, Validierung) are performed with high reliability, whereas write tasks (Korrektur, Erstellen) and Verifizierung degrade with model size and cross-layer dependency depth.
- **H2 (Tool semantics):** Exposing tool-native capabilities (validation suites, expression evaluation) as MCP tools yields larger gains than a thin CRUD wrapper over the SysML v2 API alone.
- **H3 (Model scale):** Findings obtained on small teaching models (GfSE collection) do not transfer to a large multi-layer model (Apollo 11) without a semantic navigation layer.

Approach

Use case	Example task on Apollo 11	Ground truth / metric
Abfragen (query)	“Which technical components satisfy requirement X?”	exact set match vs. model query
Validierung (validation)	“Is package P well-formed and CoSMA-conformant?”	tool validation suite results
Verifizierung (verification)	“Is mass budget requirement R satisfied by the current values?”	expression evaluation in tool
Korrektur (correction)	repair injected faults (dangling ref., wrong type, broken <code>satisfy</code>)	fault-repair rate, no collateral change
Erstellen (creation)	add missing technical component with ports + satisfy links	syntactic validity + expert rubric

Table 1: Use-case catalogue (to be refined).

[TODO 150 words prose around the table]

Evaluation Methodology

[TODO 150 words]

Work Plan

Weeks	Phase	Deliverable
1–4	Setup	Exposé; import Apollo 11 into MSoSA; MSoSA/TWC API access; literature base
5–9	Bridge	MCP server (thin tier + semantic tier); use-case task catalogue v1
10–13	Benchmark	task set with ground truth; fault injection; evaluation harness
14–16	Experiments	runs across arms and LLMs; failure analysis
17–20	Writing	thesis text, discussion, submission (19.01.2027)

Table 2: *Milestones.*

Bibliography

- [1] OBJECT MANAGEMENT GROUP: OMG Systems Modeling Language (SysML), Version 2.0 / Object Management Group. Version: September 2025. <https://www.omg.org/spec/SysML/2.0/>. 2025. – Specification. – The formal SysML v2 language specification (textual notation, KerML-based semantics). Defines the language the agent must read and write.
- [2] OBJECT MANAGEMENT GROUP: Systems Modeling API and Services, Version 1.0 / Object Management Group. Version: September 2025. <https://www.omg.org/spec/SystemsModelingAPI/1.0/>. 2025. – Specification. – Standard REST/HTTP API for SysML v2 repositories (projects, commits, branches, elements, queries). Implemented by Teamwork Cloud; candidate transport for the MCP bridge (Option A).
- [3] ANTHROPIC: *Model Context Protocol – Specification*. <https://modelcontextprotocol.io/specification>, 2025. – Open protocol for exposing tools, resources and prompts to LLM agents. The bridge to MSoSA will be implemented as an MCP server.
- [4] HELLE, Philipp ; SCHRAMM, Gerrit: Fly me to the Moon – Modeling Apollo 11 Using SysML v2. In: *Systems Engineering* (2026). <http://dx.doi.org/10.1002/sys.70074>. – DOI 10.1002/sys.70074. – [TOP 3] Presents the open-source Apollo 11 SysML v2 model (Airbus Central R&T, MPL-2.0, ≈ 7 kLOC, five CoSMA layers) explicitly intended as a benchmark for next-generation MBSE tools. Chosen as the system under test for all use-case experiments.
- [5] GFSE – GESELLSCHAFT FÜR SYSTEMS ENGINEERING: *SysML-v2-Models: A curated collection of high-quality SysML v2 models*. <https://github.com/GfSE/SysML-v2-Models>, 2025. – Curated BSD-licensed collection of small SysML v2 models, explicitly meant to help train and test LLM-based tooling. Used as secondary/generalisation test set and as source of good/bad examples.
- [6] DASSAULT SYSTÈMES: *CATIA Magic/Cameo SysML v2 Solution*. <https://docs.nomagic.com/SYSML2P/latest/>, 2026. – Vendor documentation for SysML v2 in Magic Systems of Systems Architect / Cameo: 100 % metamodel conformance, validation suites, expression evaluation, simulation, and the standard SysML v2 REST API served by Teamwork Cloud. Defines what tool-native capabilities the semantic bridge tier can expose.
- [7] AJHCS: *cameo-mcp-bridge: MCP server bridging AI assistants to CATIA Magic / Cameo*. <https://github.com/ajhcs/cameo-mcp-bridge>, 2026. – Community MCP server for Cameo (SysML v1/UML) via an embedded Java OpenAPI plugin with

162 tools. Closest engineering precedent for Option B of the bridge; no SysML v2 support and no systematic evaluation.

- [8] TODO: Using Large Language Models for System Modeling: Current State and Future Implications. In: *TODO (IEEE)*, 2025. – Structured review of 20 publications on LLMs for MBSE tasks (creation, analysis, modification, reformulation); proposes evaluation criteria and points to MCP as the emerging integration mechanism. Provides the task taxonomy the use-case catalogue builds on.
- [9] TODO: Generating SysML v2 Models from Natural Language Requirements Using Large Language Models. In: *TODO (IEEE)*, 2025. – Four-component prompting (role, ontology, few-shot, requirements) for NL→SysML v2 with precision/recall/F1 and syntax metrics. Reference point for the *Erstellen* use case and its metrics.
- [10] TODO: Mitigating Hallucinations in SysML v2 Generation Using LLMs and a Tri-Layered Knowledge Graph Reasoning Framework. In: *TODO (IEEE)*, 2025. – Three-layer KG grounding to reduce hallucinated SysML v2 constructs; measures structural validity and semantic fidelity. Motivates grounding the agent in the live model instead of a static KG.
- [11] TODO: Knowledge-Augmented Conversational Agents for Dependency-Aware SysML v2 Model Generation and Refinement. In: *TODO (IEEE)*, 2025. – SysForge multi-agent framework (designer + validator agents, dependency-aware retrieval, iterative refinement) evaluated with Pass@1/BLEU. Closest prior work for *Korrektur*; file-based and without an industrial tool.
- [12] TODO: Enabling Humans and AI Systems to Retrieve Information from System Architectures in Model-Based Systems Engineering. In: *TODO (2025)*. – GraphRAG multi-agent system over a Neo4j graph parsed from SysML v2 text (RFLP schema), 93 % retrieval accuracy on a BEV model. Reference for the *Abfragen* use case; the thesis queries the live tool instead of a derived graph.
- [13] SOFTWARE ENGINEERING INSTITUTE, CARNEGIE MELLON UNIVERSITY: *Native AI Integration for Model-Based Systems Engineering: Three Layers that Make It Work.* <https://www.sei.cmu.edu/blog/native-ai-integration-for-model-based-systems-engineering-three-layers-2026>. – [TOP 3] Controlled three-arm SysML v2 benchmark (no tools / validation loop / MCP knowledge server + skills): iterative validation eliminates syntax errors, curated knowledge raises pattern conformance. File-based; template for the ablation design and the strongest evidence that tooling shapes agent capability.

- [14] TODO: *Benchmarking AI Agents for Hardware Design Automation via MCP Tool Calling*. arXiv:2608.26199, 2026. – [TOP 3] Methodological sibling in an adjacent domain: MCP server replicating a stateful design tool, benchmark with single-op edits, dependency chains and invalid requests, seven models with ablations. Findings on tool-description detail and context scope inform the bridge design.
- [15] TODO: *What Do CAE Simulation Agents Really Need Beyond a Generic Harness?* arXiv:2609.03718, 2026. – Finds that a generic harness with domain knowledge matches specialised agent machinery for CAE tasks. Potential counter-evidence to H2; must be engaged in the discussion.